

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	C04	Assessment:	ASSESSMENT TOOL2-INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
C04 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

Smartphone Performance Optimization

Problem: Lag occurs while switching between apps.

Analysis:

L1 Cache: Very fast; stores frequently used instructions/data.

L2 Cache: Larger than L1 and reduces access to slower memory.

L3 Cache: Shared, larger cache that reduces DRAM accesses.

DRAM: Large capacity but much slower than cache.

Proposed Solution:

Use a larger and efficient **L2/L3 cache**, keep frequently used app data in cache, and use **high-bandwidth, low-latency DRAM** for active applications.

Result: Reduced memory latency, faster app switching, and improved throughput.

2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

Cloud Server Memory Bottleneck

Problem: Database queries have high latency.

Analysis:

Cache hierarchy provides very fast access to frequently used database data.

Virtual memory paging uses disk/SSD when RAM is insufficient, causing much higher latency.

Frequent page faults significantly reduce system throughput.

Proposed Solution:

Increase **L3/cache capacity**, provide sufficient **DRAM**, optimize database caching, and reduce dependence on virtual-memory paging.

Result: Faster database queries and higher server throughput.

3. Autonomous Vehicle Data Processing

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Autonomous Vehicle Data Processing

Problem: Sensor data must be processed in real time.

Analysis:

SRAM: Extremely fast and suitable for critical real-time data.

DRAM: Large capacity and high bandwidth for sensor datasets.

MRAM: Fast, non-volatile, and energy-efficient.

Proposed Hierarchy:

CPU → SRAM Cache → MRAM → High-bandwidth DRAM → Storage

Critical sensor/control data is kept in SRAM/MRAM, while large sensor datasets are stored in DRAM.

Result: Low latency, high bandwidth, and reliable real-time processing.

4. **AI Inference Edge Device**

An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

AI Inference Edge Device

Problem: Fast model loading with low power consumption is required.

Analysis:

NAND Flash: Non-volatile and high capacity but slower.

DRAM: Fast and suitable for active AI models.

RRAM: Non-volatile, potentially energy-efficient, and suitable for emerging AI memory architectures.

Proposed Hierarchy:

AI Processor → RRAM → DRAM → NAND Flash

Frequently used model parameters can be kept in RRAM/DRAM, while complete models are stored in NAND Flash.

Result: Faster model loading, reduced energy consumption, and improved inference performance.

5. **High-Performance Gaming Console**

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

High-Performance Gaming Console

Problem: Frame drops occur while loading large scenes.

Analysis:

L3 Cache: Very fast but limited in size; useful for frequently reused game data.

DRAM: Much larger and provides high bandwidth for textures, models, and scene data.

If large assets cannot fit in cache, the system depends heavily on DRAM.

Proposed Solution:

Optimize and increase **L3 cache** where practical.

Use **high-bandwidth DRAM** for large game assets.

Use efficient prefetching to move upcoming scene data into cache/DRAM before it is needed.

Result: Faster scene loading, fewer memory stalls, and reduced frame drops.

Code of Conduct:

I ____ YUVASRI.S (192521032)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1					
Q2					
Q3					

Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis